

Qn 6	Answer	Marks
(a)(i)	Gravitational and electric	[1]
(a)(ii)	Gravitational	[1]
(a)(iii)	Magnetic, gravitational and electric [If the student has left out gravitational field in (i) and/or (iii) – only one mark penalty for whole question.]	[1]
(b)(i)	Straight line with positive gradient through origin	[1]
(b)(ii)	Maximum force shown at $\theta = 90^\circ$, zero force shown at $\theta = 0^\circ$ Reasonable sine curve with F about $\frac{1}{2}$ max at 30°	[1] [1]
(c)(i)	Face BCGF shaded	[1]
(c)(ii)	Between face BCGF and face ADHE	[1]
(c)(iii)	The potential difference between face BCGF and face ADHE sets up an <u>electric field</u> which exerts an additional force on the electrons <u>in the opposite direction</u> to the magnetic force on the electrons. The magnetic force is equal in magnitude to the electrical force, net force is zero and hence resulting in the remaining electrons moving through the conductor undeflected at steady state.	[1] [1] [1]
	Max Marks	11

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